Afak For Sciences Journal
Issn: 2507-7228 – Eissn: 2602-5345
https://www.asjp.cerist.dz/en/PresentationRevue/351
Volume: 11/ N°: 01 (2026), P 542-554

The role of modern technologies in the study of Saharan rock art (DStretch and Kalimain software)

Hassiba Safrioun
University alger2
hassiba.sefriouene@univ-alger2.dz

Abstract

Modern technology has penetrated all scientific fields, and there is no doubt that the fields of rock art and parietal art have benefited greatly from them, and like other fields of research, Saharan rock art it has undergone to new technological applications. The latter was able to add a great deal to this field of research, and even change its course with new data; this ultimately led a group of researchers to discover remarkable research findings that were completely different from previous ones.

In this context, we will try to review our personal experience in using these modern technologies, including the DStretch system and the Kalimain system, which were adopted in our recent research, and to highlight the encouraging research results, which are completely different from previous ones, in order to urge students and researchers to adopt and apply such programs, given their efficiency in achieving research results that stem from correct field data, which makes us completely avoid falling into the circle of subjectivity

Article info

Received October 26.2025
Accepted December 26.2025

Keyword:

- DStretch,
- Kalimain Program,
- Saharan rock art,
- Tassili n'Ajjer,
- hand

Introduction

The For over forty thousand years, humans in the ancient and prehistoric world have expressed themselves through art, whether engraving, drawing, or sculpture. Because these arts are so ancient, some have been erased, disappeared, or faded away. While these arts in Europe existed in the depths of caves and grottos, we also find them in the Sahara such as in Tassili. For example, most of the rock art is found on the facades of shallow shelters that have been exposed for thousands of years to the danger of natural effects such as rainwater runoff, wasp nests and wind erosion, which has caused some of these scenes and sometimes all of them to be distorted and deteriorated.

For example, most of the rock art is found on the facades of shallow shelters that have been exposed for thousands of years to the danger of natural effects such as rainwater runoff, wasp nests and wind erosion, which has caused some of these scenes and sometimes all of them to be distorted and deteriorated. Also, the Lhote expedition since the fifties of last century, which exposed the scene to washing with a sponge, caused most of the scenes to be damaged afterward, not to mention the violations of visitors and tourists to them without unfairly justification.

As specialists in the field, creating a database of information on the studied scenes of the studied area is the biggest concern for researchers. The absence or neglect of any part of it can lead to incorrect research results. Therefore, researchers today resort to the information system called DStretch. So what is DStretch?

DStretch: A successful information system for achieving a reliable information bank.

DStretch is characterized as a more efficient digital information system than Photoshop, as it surpasses the latter in terms of speed and effectiveness. It was designed by Jon Harman in 2005. (Harman2005, Mark & Billo Maestrucci; 2006 & Giannilli 2008a).

For decades, researchers in rock art have relied on classical techniques in carrying out their scientific research. Archaeological surveying was done by freehand drawing, which is 'drawing with the raised hand, Or copy by direct tracking from the photographic image, but with the development of photography, especially digital, new technologies have come to serve the research. This digital revolution has made the study easier, because it allows the required thing to be achieved in a short period of time, and it also allows the complete inventory process to be completed (even the faded or erased parts of the scene), accompanied by more accurate and objective data, with making it easier to study overlays or the matchings (when one scene is drawn on top of another), and that's without resorting to touching the scenes or actual friction with them. (Brady, Gunn, McDonald et al. 2012: 628)

The best way to create an information bank

One of the essential conditions for achieving a good objective study of rock art, whether it is an engraving or a drawing, without falling into the trap of subjectivity, is that it can never be achieved except after achieving what is known as the "information bank," which is a collection of images that deal with all the scenes related to the subject being studied or the area being studied.

AFAK for Sciences Journal مجلة آفاق للعلوم EISSN: 2602-5345 ISSN: 2507-7228

The latter will only be achieved by gathering all possible artistic scenes, and because reaching them is not that easy, due to the vastness of the desert area (and most of the time the studied area as well), and with the difficulty of the terrain, not to mention the instability of the region due to the geopolitical disturbances that the region has come to know of, from civil wars and armed terrorist groups, as a result of the splitting of governments and regimes, the entire saharian region has become extremely dangerous zones.

The effectiveness of setting up an information bank under normal conditions is never easy, especially in a climate characterized by extreme heat throughout the year, deadly flash floods in some seasons, sandstorms, and difficult pathways.

This is also one of the major obstacles that can never be ignored. With all these obstacles combined in the archaeological research of saharian art, DStretch may represent the ideal solution to achieve the desired purposes. After collecting the images in the field, we process them in the same place and then store them directly in files to be studied later. Furthermore, a student or researcher who is far from the field or unable to access it can refer to the information bank with images processed by the DStretch system without travelling thousands of kilometers in potentially impossible circumstances.

Dstretch also allows us deconstructing the scenes in the images in previously captured images (provided they are of high digital value or quality) that were previously difficult to read with the naked eye, as they became classified as "faded scenes" due to the passage of time and circumstances after the researcher deconstruring the drawn patterns (which is a necessary step in this type of study), sometimes the inventory is incorrect due to the lack of parts of the scene or scenes due to their absence as a result of the decay that has affected them and is not perceived by the naked eye, which leads to the interpretation of the scene incorrectly, incompletely, or exaggeratedly.

The DStretch system has allowed for the revival of the scene by restoring it with color gradients, but the operator's skill still plays a major role in achieving the desired results because the latter must have high efficiency in operating the program to reach what is required.

DStretch also allows us deconstructing the scenes in the images in previously captured images (provided they are of high digital value or quality) that were previously difficult to read with the naked eye, as they became classified as "faded scenes" due to the passage of time and circumstances after the researcher deconstruring the drawn patterns (which is a necessary step in this type of study), sometimes the inventory is incorrect due to the lack of parts of the scene or scenes due to their absence as a result of the decay that has affected them and is not perceived by the naked eye, which leads to the interpretation of the scene incorrectly, incompletely, or exaggeratedly.

The DStretch system has allowed for the revival of the scene by restoring it with color gradients, but the operator's skill still plays a major role in achieving the desired results because the latter must have high efficiency in operating the program to reach what is required.

AFAK for Sciences Journal مجلة آفاق للعلوم EISSN: 2602-5345 ISSN: 2507-7228

(Clogg & Diaz-Andreu 2000: 842) ; (Brady, Gunn, McDonald et al. 2012: 630, 632)

Exploiting DStretch in the Study of saharan Rock Art: Results and Applications

The DStretch system was introduced and presented for the first time to specialists in rock art in 2006 as part of the annual meeting of the Friends of Saharan Rock Art Association, during which the frenchman Louis Noël Viallet presented an unclear image of a rock scene taken in the Ouan Derbaouen area.

There, he demonstrated the specialists the effectiveness of the program, so the researchers rushed to it and most of them adopted it in their subsequent studies, which were later published in the association's journals.

The results of the research were nothing short of dazzling.

Figure 1. The photo was taken in Ouan Derbaouen by Louis-Noël Viallet

Figure 2. The same document after processing DStretch –YRE the rope is clearly visible near the flock.

The experience of using DStretch in the study of saharan rock art:

We were among the pioneers in studying rock art scenes by applying this information system, similar to what some Algerian researchers like Hachid do. During the study of the Tassili rock art scenes using the DStretch system, reading the familiar rock art scenes came with many surprises .Rock scenes can be misinterpreted if the scene has been poorly survey , and this latter can be done in two different ways**:**

The first is the fading of the scene's colors (a faded scene), the absence of a part of it, distortion, or overlays (drawing one scene on top of another leads to the overlapping of shapes and colors).

The second is the complete absence of elements from the scene it puts its owner in a cycle of error.

For example, the research team of Le Quellec, using DStretch, was able to count hundreds of artistic scenes in the Grand Sahara, even in locations previously thought by specialists to be well-known. This was achieved by reviving those faded scenes through the application of this new information system. (Le Quellec, J L, & all, 2015)

The role of the DStretch in the study of human forms

During their study of the facade bearing the famous scene 'The Great God of Saffar', the Le Quellec team, along with researcher Duquesnoy, were able to identify approximately two hundred human figures, whereas the Lhote team in the fifties had counted only about fifty figures on the same facade. This demonstrates the effectiveness of the DStretch in re-examining and counting saharan art scenes and the lack of reliance or trust in previous field studies.

(Duquesnoy F.2015:298)

Likewise, we, in turn, were able, through our study of well-known scenes and through our application of DStretch, to identify this flaw, which would lead us to meaningless research results that lack scientific credibility. Sometimes we recorded the absence of clear parts of scenes, whether human or animal characters, along with the absence of many details. The latter would be necessary for this type of study, and their absence could directly lead to erroneous research results, because objective studies are based primarily on a database, and the latter depends on all the scenes that deal with the subject, and any omission of a part of these scenes or details, even if small, may lead to incorrect scientific results.

The absence of human elements, from the Well-known scenes

It happened that some researchers in saharan rock art conducted their studies based on surveys that had been completed based on photographs taken at the site. Most of those studies came out wrong, incomplete, and sometimes suspicious in terms of the results, due to the absence of elements of the scene that were not captured by the camera lens because of their decay, or accidentally dropping it.

This mistake has been made by many, whether from the new or old group of researchers (the colonial period), although the latter collected many mistakes due to the means used at that time, such as the mission of Lhote to Tassili n'Ajjer, who ordered his assistants to pass a wet sponge over the scenes before photographing them or lifting them in order to remove the dust, but this led to their damage and in many cases their destruction.

In the study that the researcher Sansoni had conducted while studying the rock art scenes and the round heads stage, he mentioned that he had completed all the surveys based on a set of good quality images. Although his work was crowned with high-quality, accurate surveys of the highest degree at that time, today, after passing those scenes through the pioneering program, we realized that today those surveys have become a dubious scientific source, as many elements and figures have escaped, and some surveys have appeared incomplete or truncated, with the absence of many details due to their not being perceptible to the naked eye.

Likewise, in the 44th survey, we can see a group of figures in supplication or in a state of entreaty or prayer, which had been completed in a uniform color scheme of black tending towards bronze color. The entire scene was completely absent from the survey made by Sansoni, although the latter is known for his professionalism and accuracy in the work. (Sansoni U.1994, fig 44)

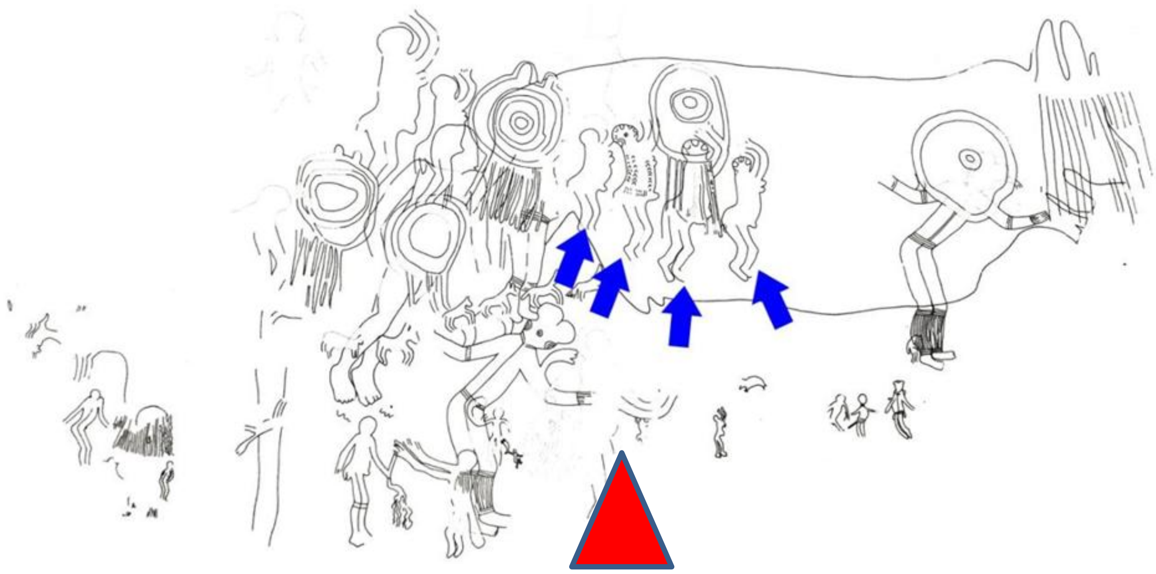


Figure 2 - The missing linear survey of the scene completed by Sansoni - the missing part is marked by the red triangle. (Sansoni U.1994, fig. 44)

AFAK for Sciences Journal مجلة آفاق للعلوم EISSN: 2602-5345 ISSN: 2507-7228

Figure 3 - The missing part of the scene of Sansoni survey

Figure 4 - The missing part of the scene Sansoni survey after processing it

with the DStretch system.

The scene of the birth in Sefar, previously studied by the Halley couple, had incomplete research results. We re-study the scene and discovered exciting details after applying this system to images we had taken in a previous mission. By deconstructing the scene, elements that had never been seen with the naked eye, the subject of the scene changed completely with the emergence of new human elements and many details that had never been seen before and would not have appeared had we not applied the DStretch system to the scene, which made us conclude in the end with research results that are completely different from those mentioned in the doctoral thesis, which made the subject take a different scientific direction. (Safrioun, H & Belkhiri L, 2020)

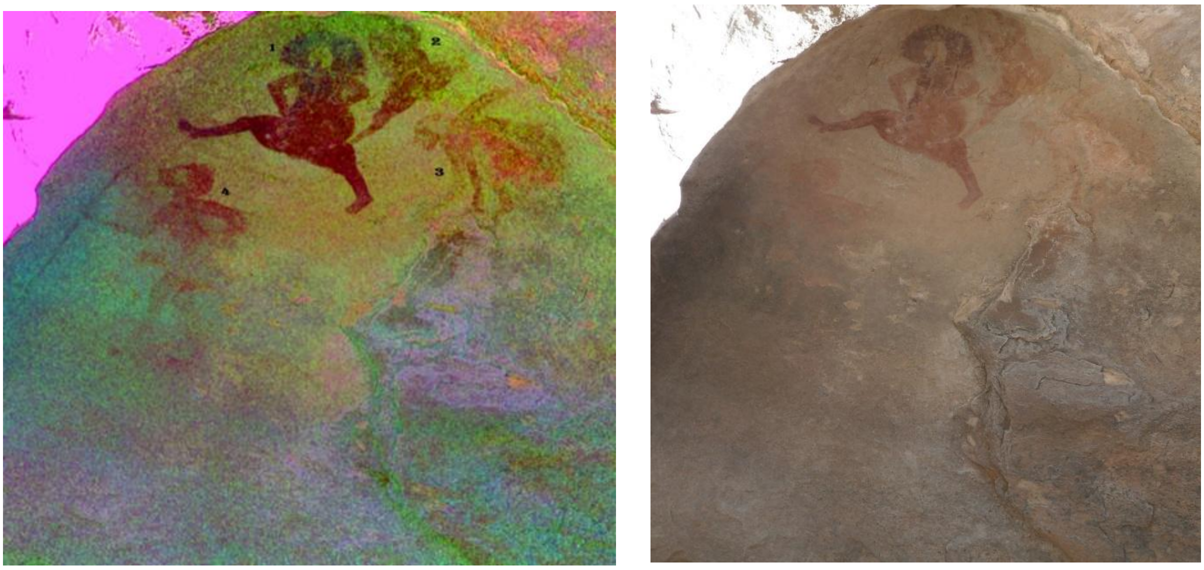


Figure 5 - Image of the birth scene before and after the treatment with DStretch (faded characters)

Identifying details in the human form: the example of Venus Tamrit

In studying human figures in saharan rock art, the details become very important, especially in specialized thematic studies such as the study of human representations, female representations, or jewelry representations, etc.

Because the state of the drawings cannot always be good, details were often incomplete or missing, which ultimately led to erroneous descriptive results, and consequently, the research results were also wrong.

One example we can browse it in this case is the Venus of Tamrit, a scene to a famous painting located in Tamrit on the Tassili N'Ajjer plateau.

The image of the Venus of Tamrit, after being processed with the DStretch system, showed that the fatty distributions carried by both female figures prove their belonging to the group of females who practice the process of pregnancy and childbirth through the distribution of fat in certain parts of the body and in that way, while it appeared in a completely different way in that lifting carried out by the French scientific mission of the researcher Lhote.

The distribution of fat on the body has significant implications. If the shape of its distribution is related to the female sex, we can read her approximate age, her reproductive function (birthing - once or repeatedly), and her function as a cover (practicing breastfeeding).

The distribution of fat in the case of Venus Tamrit indicates that the two figures are not obese, but rather are the result of practicing the maternal function. Such details are very important in this type of study, and we would not have reached them if we had relied, for example, on the lifting that was achieved by the Lhote expedition in the fifties and was studied and published later with the researcher Breuil.

AFAK for Sciences Journal مجلة آفاق للعلوم EISSN: 2602-5345 ISSN: 2507-7228

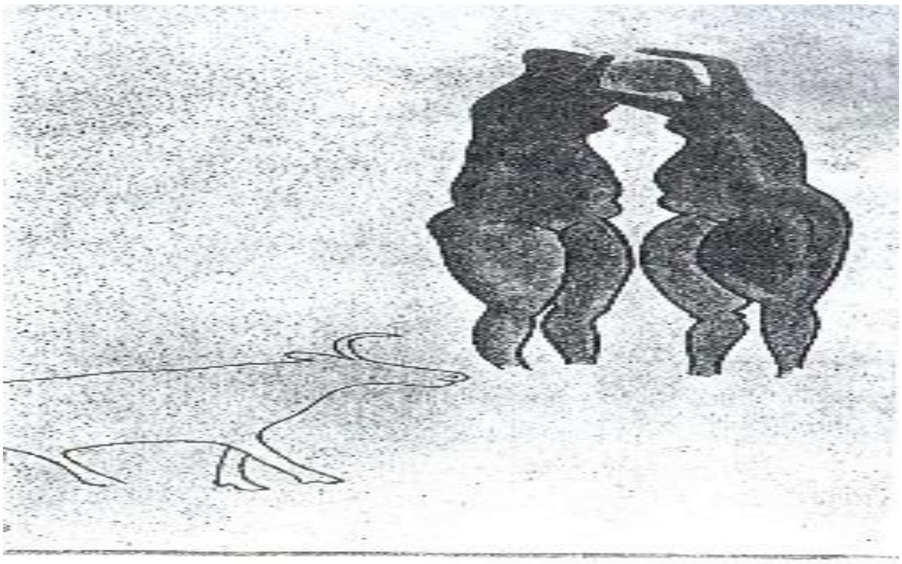


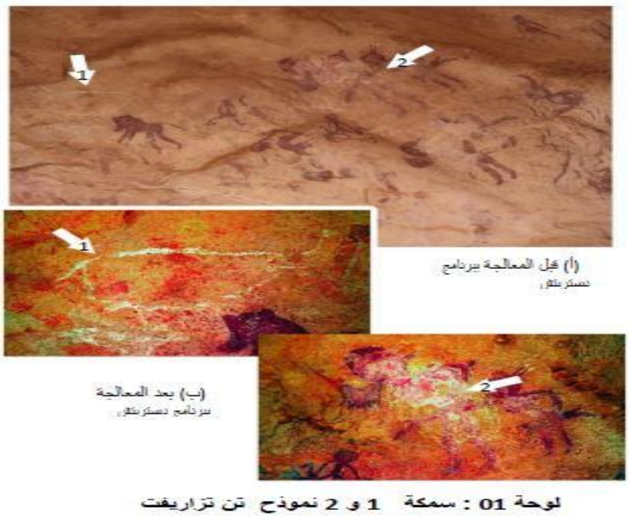
Figure 6. – A linear survey of the Venus Tamrit scene by Lhote (Lhote, H.1973:34)

Figure 7 – Venus Tamrit before the treatment with Dstretch

Figure 8 - VenusTamrit after the treatment with DStretch

Discovery of animal forms in studied scenes:

In a previous study, we used this technology to study scenes related to fishing and fish, specifically drawings from the Tassili N'Ajjer plateau. Our application of this system yielded good results, revealing new elements in previously studied scenes, including new fish forms and special traps. By applying the program to known and studied scenes, we were able to identify entirely new fish forms that had never been previously mentioned. (Safrioun H & Amari N, 2025)



b) After the treatment with DStretch

a) Before the treatment with DStretch

Finally, based on our experience and the findings of our expert colleagues such as Le Quellec and Duquesnoy, we will join their voices in urging researchers to use DStretch, it represents the most suitable information system for studying desert rock art, especially thematic studies, and particularly drawing, as research has shown. Under current circumstances, image processing (provided the images are of high quality) is essential.

One of the advantages of digital photography is the ability to collect images in a short period of time and in locations that researchers might not be able to return to, then it is processed later, so the process of inventorying the scenes becomes faster and more credible, which allows for the creation of an information bank or a digital inventory of scenes and sites, which facilitates the process of exchanging it between specialists and responsible research organizations with the least effort and costs.

Results of applying the Kalimain program to saharan rock art

Among the earliest representations that ancient man tried to document on walls or rock facades was the representation of the hand (palm print), a theme that would be shared by Paleolithic and Neolithic artists in all ancient artistic regions (Europe, Asia and Africa).

We say that the representation of the hand (palm print) is negative when the artist sprays a colored substance on the hand placed on top of the facade, and when it is lifted, it will leave its mark in the middle of the colored halo. The representation of the negative hand (palm print) is more common than the positive hand (palm print). The latter is done after dipping the hand (palm print) in a colored substance and imprinting it on the facade, so it leaves a colored mark.

This artistic representation is in fact the most reliable link that connects modern man with ancient man.

Kalimain's program for determining the sex of a negative hand:

Researchers Michel Chazine and Arnaud Noury were able to create a pioneering software program to determine the gender of the negative hand (palm) depicted in rock art. By applying it to a picture of a negative hand palm (the picture must be of high quality), it is possible to determine the gender of the owner of the hand palm, whether male or female!

The idea for the program was inspired by an idea from two researchers specializing in fingerprints, Kevin J. Sharpe and Leslie Van Gelder, which stated that the gender of a negative drawn palm can be determined by using the Manning Index, which states that the ratio of length between the index and ring fingers represents the gender identity of each individual, being equal in females and different in males, in the early months of foetal development, certain hormones influence finger growth; estrogen affects the growth of the index finger, while testosterone affects the growth of the ring finger and from this standpoint Shazine's idea came. Knowing the gender of the hand palm or hand being drawn may enable us to determine the gender of the artist, whether male or female.

AFAK for Sciences Journal مجلة آفاق للعلوم EISSN: 2602-5345 ISSN: 2507-7228

Although the use of the Manning index in this type of study has been subject to much controversy and contradictions, it may ultimately lead to giving an idea about the gender of the artist, as indicated by the preliminary study that we are currently conducting with the French technician Arnaud Noury, which may answer us about the origin of the gender of the Tassili artist, whether he was male or female? It is a question we have been trying to answer for decades, after serious research and suffering.

Prospects for using modern technologies in rock art studies

The DStretch and Kalimain programs are among the most promising information programs for studying desert rock art, as proven by our attempt to utilize them in our last studies. The effectiveness of DStretch has enabled us to achieve a sound and complete information bank that can be adopted by specialists as a sound basis for achieving and completing their future research. Our initial application of the Kalimain program also yielded promising results we can later utilize this information to study the group of hand palm and hands and depicted in saharan rock art, reinterpreting them from the perspective of the division of artistic tasks between males and females, perhaps even assigning the latter a new position within this division. It remains to be seen that in the future we will not be able to conduct serious research except by applying these new and innovative information systems and programs.

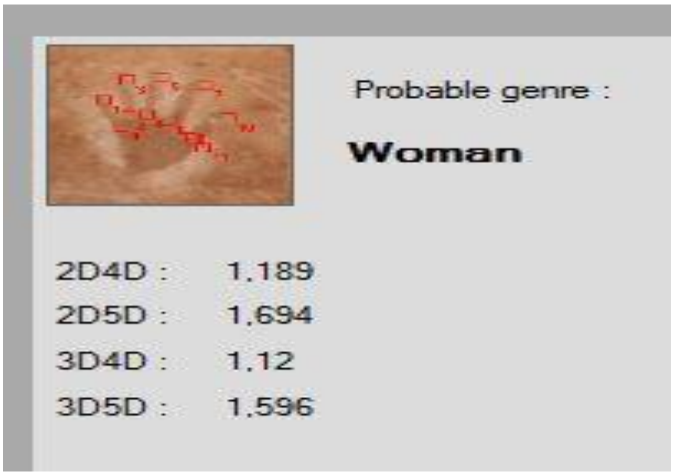


Figure 10 - Application of the Kalimain program to a negative of print image of a scene from Tin Tazarift belonging to the female gender. (The program application has done by Arnaud Noury)

Bibliography

Brady Liam M. 2006. Documenting and Analysing Rock Paintings from Torres Strait, NE Australia, with Digi- tal Photography and Computer Image Enhancement. *Journal of Field Archaeology* 31(4): 363-379.

Brady Liam M., Robert G. Gunn, Jo McDonald & Peter Veth 2012. Digital Enhancement of Deteriorated and Superimposed Pigment Art: Methods and Case Stu- dies *In* Jo McDonald & Peter Veth, *A companion to rock art* (pp. 627-643). Oxford: Blackwell.

Clogg Phil & Margarita Diaz-Andreu 2000. Digital Image Processing and the Recording of Rock Art. *Journal of Archaeological Science* 27: 837–843.

DUQUESNOY F. 2015. *Apport des outils numériques et informatiques à l'étude des images rupestres du Sahara central : exemple d'application aux peintures de Séfar (Tasīli-n-Ǎjjer), Algérie.* Aix-en-Provence, Aix-Marseille Université (thèse de doctorat).

Hachid Malika, Jean-Loïc Le Quellec, Ali Amara, Lucile Beck, Évelyne Kaltnecker, Souhila Merzoug, Anita Quiles & Hélène Valladas 2012. Quelques résultats du projet de datation directe et indirecte de l'art rupestre saharien. *In The Signs of Which Times? Chronological and Palaeoenvironmental Issues in the Rock Art of Northern Africa* (pp. 71-94). Bruxelles: Royal Academy for Overseas Sciences.

Harman Jon 2005. *Using Decorrelation Stretch to Enhance Rock Art Images.* http: //www.petrogly- phs.us/article_using_decorrelation_stretch_to_ enhance_rock_art_images.htm/, 14 November 2013.

Maestrucci Fabio & Gianna Giannelli 2008a. La foto- grafia nella documentazione dell'arte rupestre: Tec- niche e metodi del rilievo fotografico. *Les Cahiers de l'AARS* (12): 221-231

LE QUELLEC J.-L., DUQUESNOY F. & DEFRASNE C. 2015. « Digital image enhancement with DStretch®: Is complexity always necessary for efficiency? », *Digital Applications in Archaeology and Cultural Heritage*, 2-2/3: 55-67. https://doi.org/10.1016/j.daach.2015.01.003. DOI: 10.1016/j.daach.2015.01.003

Louiza Belkhiri, & Hassiba Safrioun. (2024). About Discrepancies In The Rock Paintings Of Tassili N'ajjer Between Taken Paintings Statements And Originals. Educational Administration: Theory and Practice, 30(7), 287–296. https://doi.org/10.53555/kuey.v30i7.6658

Safrioun H & Belkhiri L 2020: *The Postures of Childbirth in the Bovidian Women in the Rock Art of Tassili N'ajjer, Central Sahara of Algeria* in *Anthropomorphic Images,* Ed, Archaeopress Archaeology Oxford

Safrioun H. Nabilla A. (2025). *Discovery of New Fish Depictions in the Central Sahara (Tassili n'Ajjer Plateau – Algeria).* Science, Education and Innovations in the Context of Modern Problems, 8(5), 693-705; doi:10.56352/sei/8.5.71. https://imcra-az.org/archive/363-science-education-and-innovations-in-the-context-of-modern-Problems-issue-5-volviii-2025.html

AFAK for Sciences Journal مجلة آفاق للعلوم EISSN: 2602-5345 ISSN: 2507-7228